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BATCH: CSE A-2

TITLE : Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Problem Statement

Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.

Learning Objectives

To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.

Theory

Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

```
package Exp14;

public class Main1 {
    Run | Debug
    public static void main (String[] args) {
        int num1 = 10;
        int num2 = 0;

        try {
            int result = num1 / num2;
            System.out.println("Result : " + result);
        } catch (ArithmeticException e) {
            System.out.println(x: "Error : Cannot divide by zero.");
            System.out.println("Error : " + e.getMessage());
        }

        System.out.println(x: "Program continues after exception handling.");
    }
}
```

```
package Exp14;

public class Main2 {
    Run | Debug
    public static void main (String[] args) {
        try {
            String str = "123a";
            int num = Integer.parseInt(str);

            int r = 10 / 0;
        } catch (NumberFormatException | ArithmeticException e) {
            System.out.println("Error : " + e.getClass().getSimpleName() + " occurred.");
        }
    }
}
```

```

package Exp14;

public class Main3 {
    Run | Debug
    public static void main (String[] args) {
        try {
            System.out.println(x: "Outer try block started");

            try {
                System.out.println(x: "Inner try block started");
                int result = 10 / 0;
            } catch (ArithmeticException e) {
                System.out.println("Error + " + e.getMessage());
            }

            String str = null;
            System.out.println(str.length());
        } catch (NullPointerException e) {
            System.out.println("Caught NullPointerException in outer try block : " + e.getMessage());
        } finally {
            System.out.println(x: "Finally block of outer try executed");
        }

        System.out.println(x: "Program continues after nested try-catch blocks");
    }
}

```

OUTPUT

```

PS D:\Java-Questions\Exp14> java Main1.java
Error : Cannot divide by zero.
Error : / by zero
Program continues after exception handling.
PS D:\Java-Questions\Exp14> java Main2.java
Error : NumberFormatException occurred.
PS D:\Java-Questions\Exp14> java Main3.java
Outer try block started
Inner try block started
Error + / by zero
Caught NullPointerException in outer try block : Cannot invoke "String.length()" because "<local1>" is null
Finally block of outer try executed
Program continues after nested try-catch blocks

```

Exercise

1. Create an ATM program handling invalid withdrawal amount using try-catch

```
// Create an ATM program handling invalid withdrawal amount using try-catch.

import java.util.Scanner;

public class Exp1 {
    Run | Debug
    public static void main(String[] args) {
        try (Scanner in = new Scanner(System.in)) {
            double balance = 50000;

            try {
                System.out.print(s: "Enter withdrawal amount: ");
                double amount = in.nextDouble();

                if (amount <= 0) {
                    throw new Exception(message: "Invalid withdrawal amount!");
                }

                if (amount > balance) {
                    throw new Exception(message: "Insufficient balance!");
                }

                balance -= amount;

                System.out.println(x: "Withdrawal Successful.");
                System.out.println("Remaining Balance: Rs." + balance);
            } catch (Exception e) {
                System.out.println("Error: " + e.getMessage());
            }
        }
    }
}
```

```
PS D:\Java-Questions\Exp14\Exercise> java Exp1.java
Enter withdrawal amount: 4000
Withdrawal Successful.
Remaining Balance: Rs.46000.0
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

```
// Create an Online Shopping program that handles an invalid product quantity entered by the u
// Display an appropriate error message if the quantity is less than or equal to zero.

import java.util.Scanner;

public class Exp2 {
    Run | Debug
    public static void main (String[] args) {
        int products = 10;

        try (Scanner in = new Scanner(System.in)) {
            try {
                System.out.print(s: "Enter number of products to but : ");
                int am = in.nextInt();

                if (am < 0) {
                    throw new Exception(message: "Invalid Product Amount.");
                }

                if (am == 0) {
                    throw new Exception(message: "Please enter amount more than 0.");
                }

                if (am > products) {
                    throw new Exception(message: "Not enough products in stock to buy.");
                }

                System.out.println(x: "Purchased!");
                products -= am;
                System.out.println("Remaining Products : " + products);
            } catch (Exception e) {
                System.out.println("Error: " + e.getMessage());
            }
        }
    }
}
```

```
PS D:\Java-Questions\Exp14\Exercise> java Exp2.java
Enter number of products to buy : 4
Purchased!
Remaining Products : 6
PS D:\Java-Questions\Exp14\Exercise> |
```